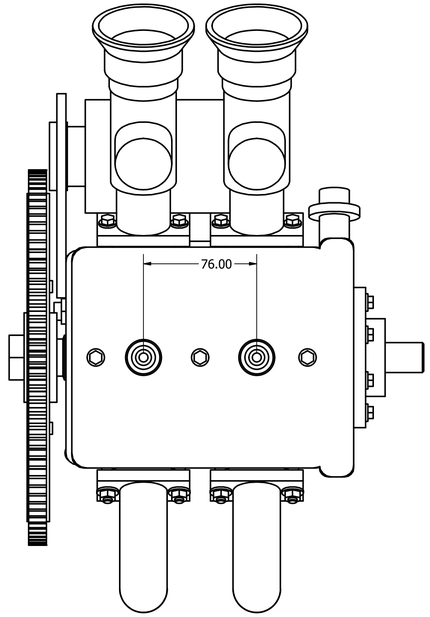
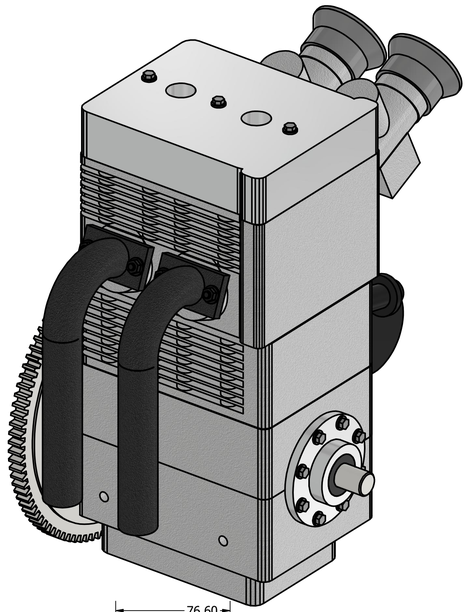
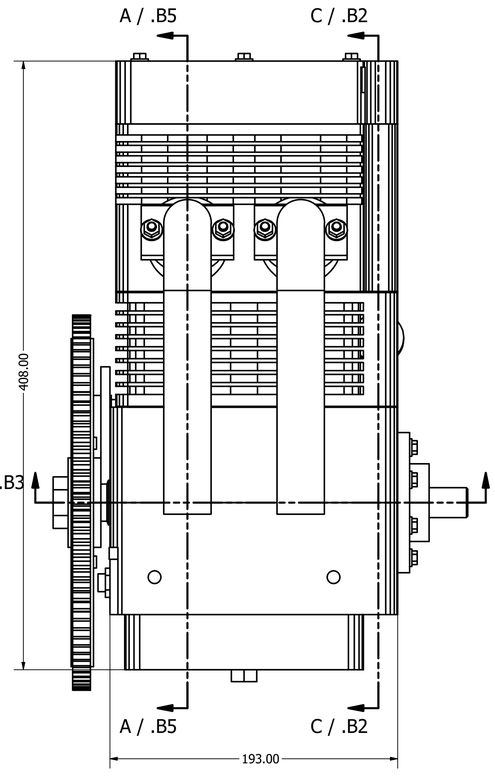
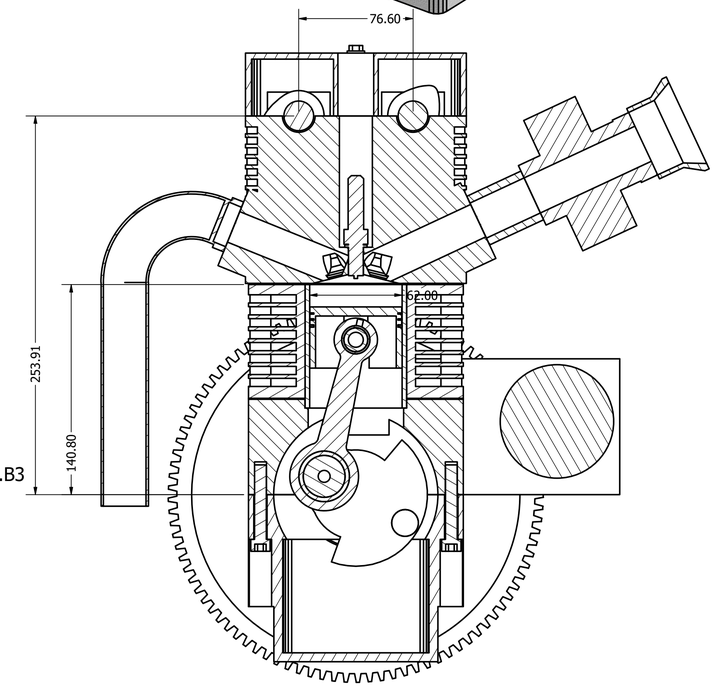
TWOCYLINDERENGINE - 296 CC AIR-COOLED 4-STROKE PARALLEL TWIN
(DESIGNED AND CALCULATED, NOT BUILT)
BORE x STROKE 62 x 49 (EX300 PISTONS, RINGS, SPRINGS, BUCKETS), ROD 94
CR 10.6 (CHAMBER 12.01 CC), 180 DEG CRANK, FIRING AT 180 / 540
DOHC 4 VALVES / CYL (CUSTOM 22 / 19 HEADS), DIRECT BUCKETS 25
CAMS 7.15 LIFT: IVO 14 BTDC / IVC 46 ABDC, EVO 48 BBDC / EVC 12 ATDC
PRESSED CRANK ON 2 x 6306/C3 (PINS PRESS 0.065-0.082), RODS ON K28x34x17
BALANCE FACTOR 0.44 (W-ALLOY SLUGS), NO BALANCE SHAFT
CALC.: 20.9 KW AT 8500 RPM, 24.4 NM AT 6500 RPM
VALVE FLOAT 9967 RPM; MIN. PISTON-VALVE 2.05 MM
ASSEMBLY PARAMETER Crank (0 = CYL 1 TDC FIRING); SHOWN AT 60 DEG
ENVELOPE 278 x 464 x 473 MM; 36.7 KG (INVENTOR)
DIMENSIONS: DESIGN VALUES, EACH CHECKED AGAINST THE MODEL

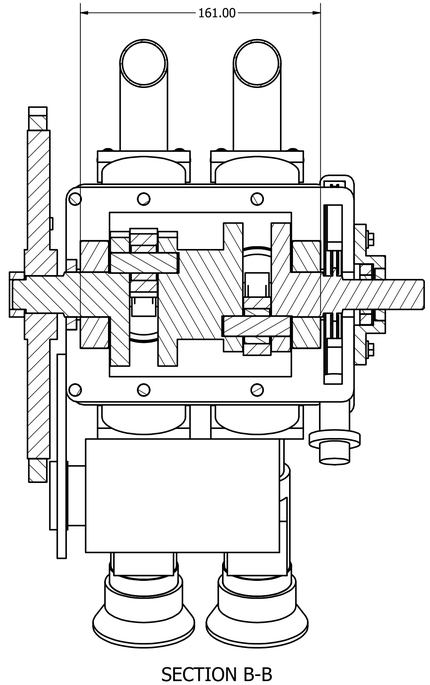






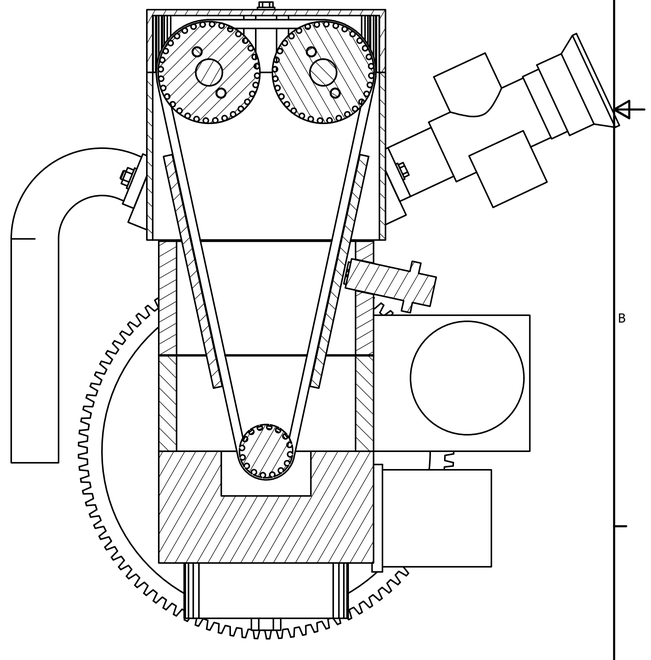


SECTION A-A
SCALE 1 / 2 / .B7

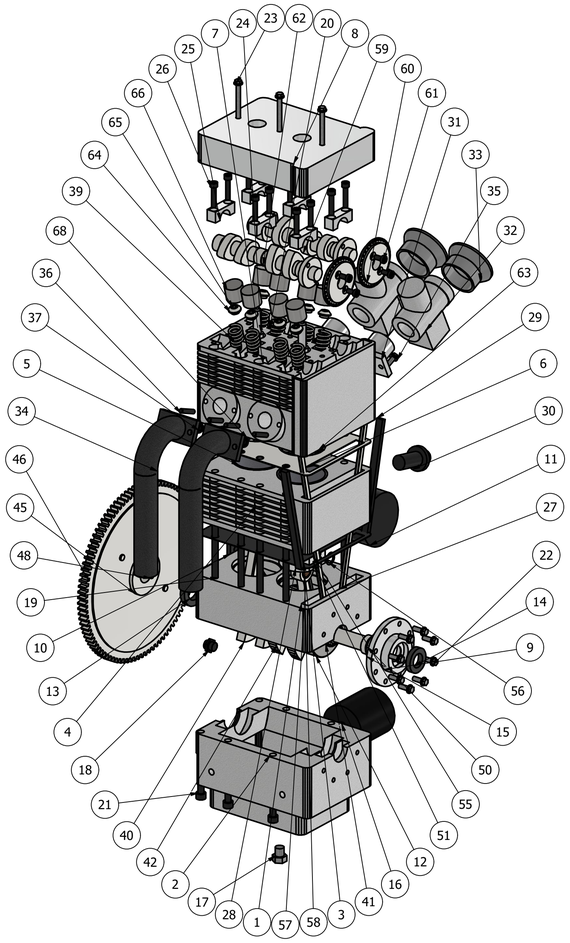


SECTION B-B
SCALE 1 / 2 / .B7

SECTION C-C
SCALE 1 / 2 / .B7



DRAWN Jay Patel	11/2/2026		
CHECKED		TITLE	
QA		TwoCylinderEngine - general assembly	
MFG			
APPROVED			
		SIZE D	DWG NO TCE-DWG-001 REV
		SCALE 1 / 2	SHEET 1 OF 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
53	2	TCE-203	Second ring 62 x 0.8, taper face (kit)
54	2	TCE-204	Oil control ring 62 x 1.5, 3-piece (kit; modelled as one)
55	2	TCE-205	Piston pin 16 x 44.5, case-hardened (Kawasaki 13002-0710)
56	4	TCE-206	Piston-pin circlip 16 mm (Kawasaki 92033-0732)
57	2	TCE-106	Connecting rod, one piece, 4340 QT: 94 centres, big end 34 bore x 18.4 (K28x34x17 needles, bore carburised + ground as the outer race), small end 19 bore for a 16x19 bronze bush, I-beam 12 thick
58	2	TCE-107	Small-end bush, bearing bronze 16 x 19 x 16 (pressed in, reamed to 16.02-16.04 for the pin)
59	1	TCE-301	Camshaft, intake: 4140 nitrided, 4 lobes base R17, lift 7.15 (polynomial, +-68 cam deg), nose 233, 323 deg, journals 22 x 3
60	2	TCE-303	Cam sprocket 32T ANSI 25 (PD 64.78), 1045, 2 x M6 on PCD 32 (slotted for cam timing)
61	4	TCE-710	Low head cap screw DIN 7984 M6 x 12 class 10.9 (cam sprockets, Loctite 243)
62	1	TCE-302	Camshaft, exhaust: 4140 nitrided, 4 lobes base R17, lift 7.15 (polynomial, +-68 cam deg), nose 126, 216 deg, journals 22 x 3
63	4	TCE-304	Intake valve 22 head, 4.5 stem, 89 long: custom valve (EX300 4.5 stem, keeper groove and length; smaller head than the EX300 so the in / ex seats keep a 2.3 mm bridge)
64	8	TCE-307	Valve spring retainer (EX300 OEM pattern)
65	8	TCE-308	Valve keepers (cotters), pair, 4.5 stem, single groove (EX300 OEM pattern)
66	8	TCE-309	Valve lifter bucket 25 OD (direct acting, shim under bucket; EX300 OEM pattern)
67	8	TCE-310	Valve adjusting shim 7.48 x 2.00 nominal (under bucket; final sizes from the lash check)
68	4	TCE-305	Exhaust valve 19 head, 4.5 stem, 89 long: custom valve (EX300 4.5 stem, keeper groove and length; smaller head than the EX300 so the in / ex seats keep a 2.3 mm bridge)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
27	1	TCE-312	Cam chain ANSI 25H (1/4 in pitch), 118 links endless (pitch line 748.9 mm = 117.94 pitches; tensioner takes up the rest); modelled as its two side-plate bands, rollers / pins not modelled
28	1	TCE-313	Cam-chain fixed guide, tight (front) span: PA46-faced steel blade 10 wide, 0.4 mm off the chain (pinned in the tunnel)
29	1	TCE-314	Cam-chain tensioner blade, slack (rear) span, PA46-faced steel, pivot at the lower end (shown at the nominal path)
30	1	TCE-315	Cam-chain tensioner, automatic (Kawasaki EX300 12048-0084), envelope; plunger on the rear blade
31	2	TCE-501	Intake manifold 6061: flange 62 x 36 x 8 (2 x M6 on 48 centres), 36 spigot for the VM26 clamp, 26 bore; O-ring face to the head
32	2	TCE-605	Carburettor Mikuni VM26-606 round slide, 26 bore (envelope)
33	2	TCE-502	Velocity stack / filter adapter, printed ASA: slips on the VM26 air spigot (46), 68 bell for a 48-60 pod filter
34	2	TCE-503	Exhaust header 1.25 in x 0.063 mild steel, 45 mm CLR 112 deg mandrel bend to a 150 down pipe, flange 62 x 36 x 8 (2 x M6 studs); slip-on muffler TCE-631 not modelled
35	4	TCE-707	Hex flange bolt M6 x 22 class 8.8 (intake manifolds)
36	4	TCE-708	Stud M6 x 30 class 8.8 (exhaust flanges; 14 in the head)
37	4	TCE-709	Flange nut M6, brass-plated steel (exhaust)
38	8	TCE-311	Valve guide 9.5 OD x 4.52 ID x 38 (EX300 OEM pattern), pressed, 0.03 interference
39	8	TCE-306	Valve spring 19.5 OD x 2.9 wire, installed 32 (single, EX300 OEM pattern; rate and loads in the calc report)
40	1	TCE-101	Crank web, left, with mainshaft: 4340 QT, web 13 thick, hub R30 / pin boss R21 / counterweight R48 x 160 deg, crankpin bore 28 (bore 27.915 +/-0.004: press 0.065-0.082), W-alloy slug bore 18, 30 k5 bearing + seal journal, 25 flywheel seat with key 8, M20x1.5 nut thread
41	1	TCE-102	Crank web, right, with mainshaft: 4340 QT, web 13, counterweight R48 x 160 deg, crankpin bore 28 (press), 30 k5 6306 seat, integral 16T ANSI 25 cam-chain sprocket (PD 32.55, teeth induction hardened), 20 pump drive slot, 20 h6 seal + output journal with key 6
42	1	TCE-103	Crank centre section, one piece 4340 QT: two webs 13 thick (180 deg) joined by an R30 hub, counterweights R48, 2 crankpin bores 28 (bore 27.915 +/-0.004: press 0.065-0.082)
43	2	TCE-104	Crankpin 28 x 45, 52100 hardened 60 HRC, ground (needle race), 8 mm bore; pressed into the webs (0.065-0.082 interference, slip SF 2.1)
44	2	TCE-603	Needle roller and cage assembly INA K28x34x17 (big end; envelope)
45	1	TCE-108	Flywheel 1045: hub 60 (25 bore + key 8), disc 15 to the ring-gear seat, 2 trigger tabs at r 62 180 deg apart (VR pickup); carries the shrunk-on ring gear TCE-110
46	1	TCE-110	Starter ring gear 97T 10 DP (PD 246.4, OD 251.5, ID 220), Kohler Command V-twin pattern, shrink-fit on the flywheel; teeth shown as trapezoids
47	4	TCE-112	Counterweight slug, tungsten heavy alloy (W-Ni-Fe, 17 g/cc) 18 x 13, pressed + staked in each web at r 38 (4 off: raises the reciprocating balance factor from about 0.1 to about 0.45)
48	1	TCE-712	Round key: dowel pin 8 h6 x 20 in half-round seats (flywheel); cam sprocket uses the 8 x 10
49	1	TCE-711	Hex thin nut DIN 439 M20x1.5 class 8 (flywheel, 120 N m + Loctite 270)
50	1	TCE-111	Gerotor inner rotor, 4 lobes, 26 tip dia x 8, on the crank nose (drive pin 6 not modelled); wire-EDM 4140 nitrided
51	2	TCE-201	Piston 62.00 (Kawasaki EX300 pattern; envelope: compression height 22, pin 16, skirt 61.96, rings 0.8 / 0.8 / 1.5); flat crown modelled
52	2	TCE-202	Top compression ring 62 x 0.8 (kit)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	TCE-401	Upper crankcase, horizontally split at the crank axis: crank chamber R55, 2 x 6306 half seats (line-bored with the lower case), 68.2 liner spigot bores, 8 x M8 stud threads, cam-chain cavity, 30x47x7 seal bore; one-off from billet
2	1	TCE-402	Lower crankcase + wet sump (about 1.8 L): 2 x 6306 half seats line-bored with TCE-401, 6 x M8 case bolts from below, spin-on filter boss (M20x1.5) on the rear, M12x1.5 drain, 4 x M10 test-stand mount threads; one-off from billet
3	1	TCE-410	Base gasket, 0.5 fibre sheet, cut in-house (thickness also sets the deck clearance)
4	1	TCE-403	Cylinder block, air-cooled: 2 x 68 H7 liner bores on 76 centres, 7 fins 2.5 at 8 pitch, 8 stud bosses, cam-chain tunnel with tensioner bore; one-off from billet
5	2	TCE-404	Cylinder liner, centrifugally cast iron, finished 62.00 H6 bore, 68 OD press fit, 82.8 long (machined from a sleeve blank)
6	1	TCE-409	Head gasket MLS 0.8 compressed, 63 fire rings (custom laser-cut; Cometic custom service class)
7	1	TCE-405	Cylinder head, air-cooled DOHC 4-valve: pent roof 2 x 14 deg (chamber 12.01 cc), 8 x 25 bucket bores, guide bores 9.5, 2 x M10x1 plug wells, ports 24 / 22 with flanged bosses, cam line bores 22 (with caps), 8 stud C'bores, 7 fins front and back; one-off from billet, valve seats cut for sintered inserts
8	1	TCE-407	Cam cover 6061 shell 4 mm with chain-case bulge, 3 x M6 shouldered bolts on rubber grommets, 2 plug-cap holes 24
9	1	TCE-408	Right cover 6061: 94 flange, 8 x M6 on PCD 80, gerotor pump pocket 41 (pump inlet / outlet drillings not modelled), 20x35x7 output seal
10	1	TCE-411	Starter bracket, 6 mm S355 plate: holds the starter nose (42 bore, 2 x M8 flange bolts not modelled) and bolts to the case left face (2 x M8)
11	1	TCE-607	12 V Bendix starter, Kohler Command pattern (12 098 22-S class), 16T 10 DP pinion assumed (confirm); envelope
12	2	TCE-601	Deep groove ball bearing 6306/C3 30 x 72 x 19 (outer mains; C 29.6 kN)
13	1	TCE-609	Radial shaft seal 30 x 47 x 7 TC NBR (left main)
14	1	TCE-611	Radial shaft seal 20 x 35 x 7 TC NBR (output)
15	1	TCE-412	Gerotor outer rotor, 5 lobes, 40.6 OD x 8 (turns eccentric at 4/5 crank speed in the cover pocket; shown concentric as an envelope); wire-EDM 4140 nitrided
16	1	TCE-606	Spin-on oil filter Hiflofiltro HF303 (M20x1.5 thread; 65 OD assumed, 73 tall)
17	1	TCE-714	Oil drain plug M12x1.5 + crush washer (Kawasaki 92066-0767 + 92022-1827 pattern)
18	1	TCE-608	VR crank-trigger pickup M12 (2 tabs on the flywheel, one per cylinder) for the DC-CDI
19	8	TCE-701	Head stud M8 x 166 class 10.9 (8 off; 20 into the upper case, through block and head)
20	8	TCE-702	Hex nut ISO 4032 M8 class 10 + hardened washer (washer not modelled); head studs, torque in the calc report
21	6	TCE-704	Socket head cap screw ISO 4762 M8 x 50 class 10.9 (crankcase halves, from below)
22	8	TCE-705	Hex flange bolt M6 x 16 class 8.8 zinc (right cover)
23	3	TCE-706	Hex flange bolt M6 x 55 class 8.8 (cam cover, with rubber grommet and sleeve, not modelled)
24	2	TCE-604	Spark plug NGK CR8E (1275): M10x1.0, 19 reach, 16 hex
25	6	TCE-406	Cam bearing cap 14 x 42 x 16, 6061-T6, 2 x M6 on 31 centres (6 off, line-bored with the head, matched and numbered)
26	12	TCE-703	Socket head cap screw ISO 4762 M6 x 30 class 10.9 (cam caps, 10 N m)

DRAWN	Jay Patel	11/2/2026	
CHECKED			
QA			
MFG			
APPROVED			

TITLE: TwoCylinderEngine - general assembly

SIZE: D | DWG NO: TCE-DWG-001 | REV:

SCALE: 1 / 4 | SHEET 2 OF 2